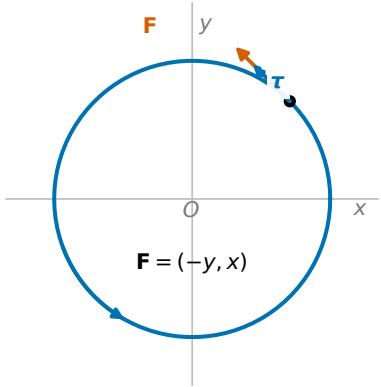
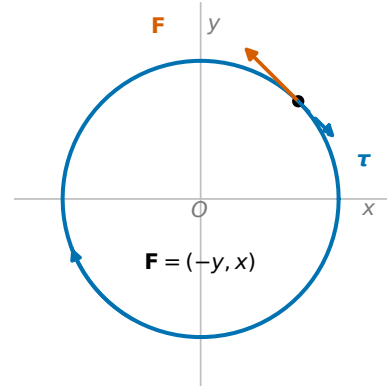
A



$\int_C ds = 2\pi \qquad \oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi$

B



$\int_C ds = 2\pi \qquad \oint_C \mathbf{F} \cdot d\mathbf{r} = -2\pi$